Optimal Defense Spending and Taxation

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Motivation

- We are experiencing a period of rising geopolitical risk.
- Beyond its human toll, realized conflict can destroy productive assets, including factories, energy facilities, and transportation infrastructure.
- Countries are responding by large increase in planned defense spending.
  - U.S. FY2027 proposal: \$1.15tn for discretionary defense (+28%); DOW +44%.
  - EU "Readiness 2030": € 800bn for additional defense, 2025–2028.
- But financing defense requires taxes that may themselves distort investment and economic activity.

Outline

1. Research Question
2. Empirical motivation
3. Model and first best
4. Decentralized economy
5. Quantitative
6. Stochastic threat

This paper asks these questions

1. How should the economy allocate resources between **productive investment** and **protective investment** (defense)?
2. How should defense be financed when the capital being protected can also be taxed?
3. How does the inability to commit affect capital taxation, investment, and defense provision?
4. Can the model account for observed defense-spending dynamics using historical geopolitical risk?

Main idea

- I develop an optimal fiscal policy model with endogenous defense.
- The economy allocates resources between
  - productive investment, which accumulates capital;
  - protective investment, which increases capital survival.
  - consumption
- If conflict occurs, part of installed capital is destroyed; defense raises the fraction that survives.
- Defense preserves productive capital, but financing it may require taxing the capital being protected.
- **Central trade-off** for gov't: tax capital to protect capital
  - Capital tax has **two opposing effects** on future productive capacity: it reduces capital, but potentially increase future capital.

Literature

- **Optimal fiscal policy and defense**
  Lucas and Stokey (1983), Aiyagari et al. (2002), Valaitis and Villa (2026).

  Valaitis and Villa studies labor taxation and debt. I focus on the dual impact on capital taxation.

- **Disaster prevention and adaptation**
  Barrage (2019), Cai and Lontzek (2019), Douenne et al. (2022), Hong et al. (2023).

  Both study an endogenous protection decision; I introduce distortionary financing and limited commitment into the protection problem

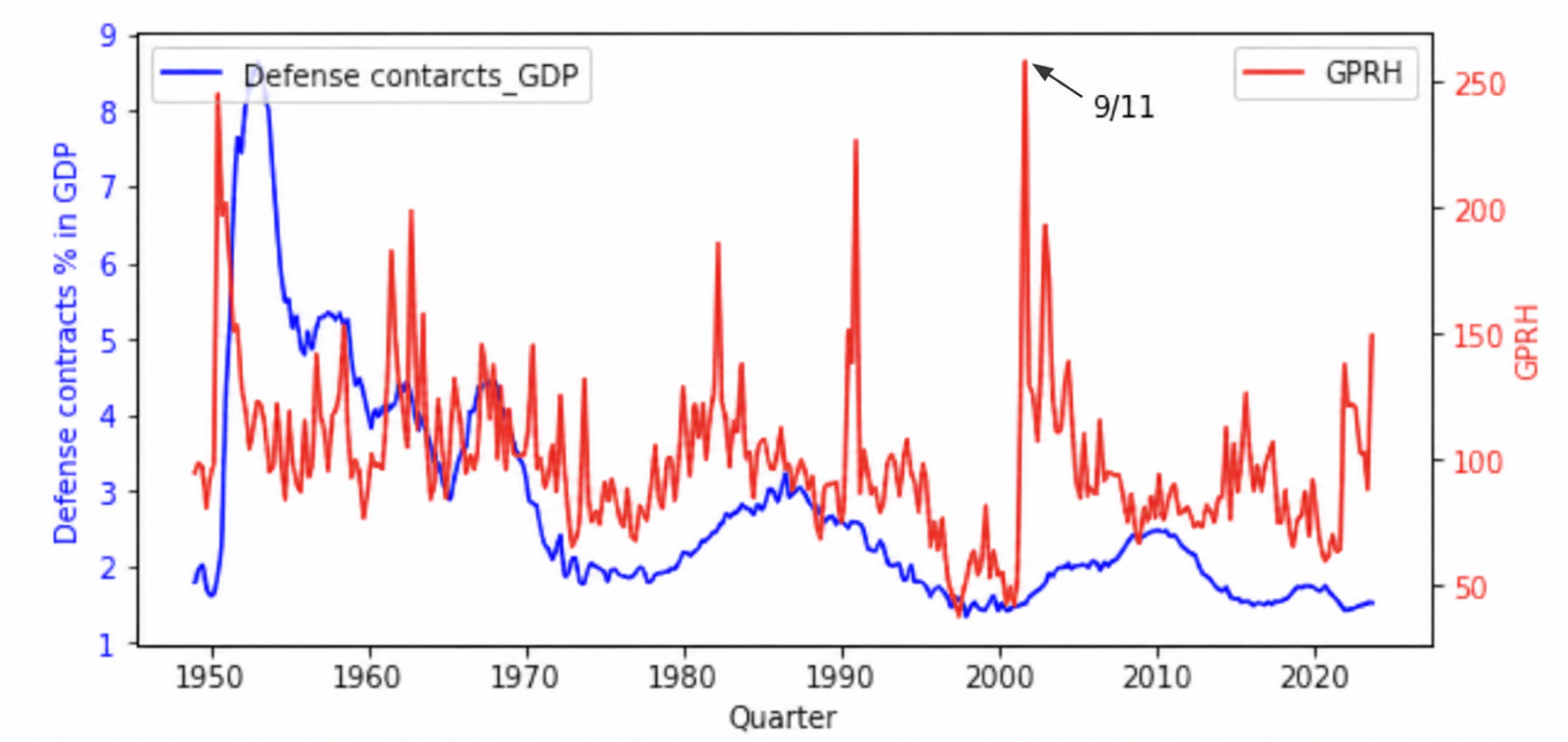
- **Macroeconomic effects of conflict**
  Pflueger and Yared (2024), Levine and Ohanian (2024), Federle et al. (2025).

  They focus on effects on output, capital, investment etc. I focus on optimal response to geo risks and how to finance it.

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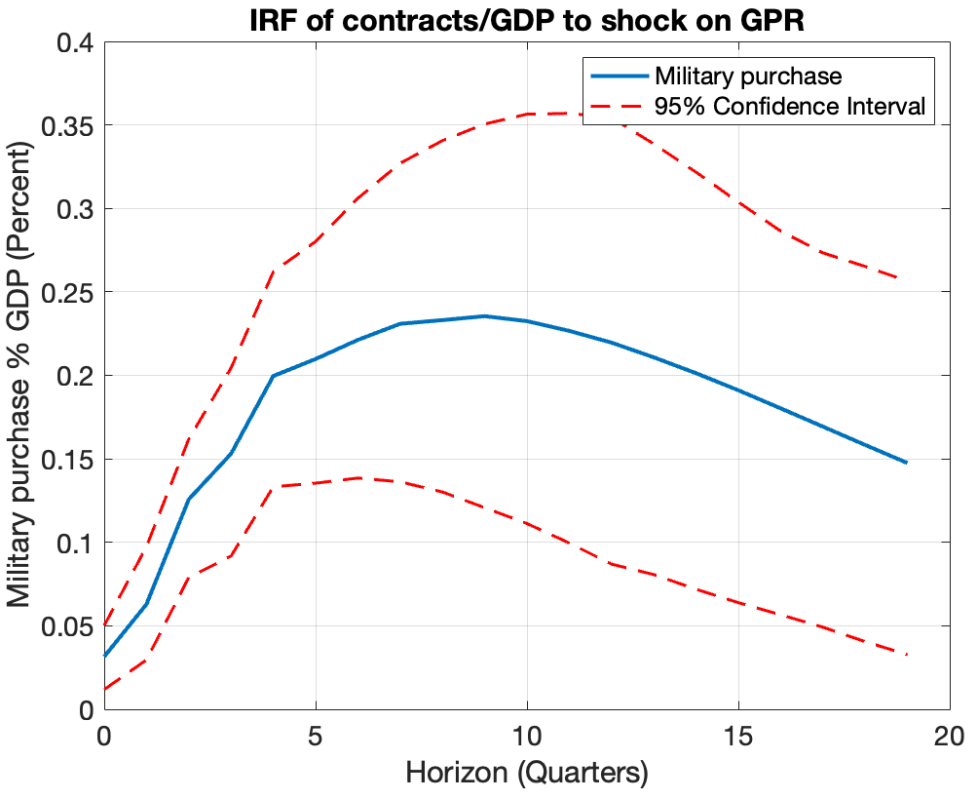
Geopolitical risk and U.S. military purchases



The Geopolitical Risk (GPR) Index (Caldara & Iacoviello 2022, AER) is measured by the frequency of newspaper articles discussing adverse geopolitical events. The U.S. as an example. Periods of elevated geopolitical risk **coincide** with greater military purchases.

▸ Decomposition ▸ Variation

The defense response is large and persistent



Response to a one SD GPR shock

- Large and persistent response
- 0.24 ppt of GDP at peak after two years.
- eqv. \$50 bn per year, remains elevated for nearly five years.

Data implies that defense is an endogenous policy response to threat.

Question left open: How should defense, allocations, and taxes respond to geopolitical threat?

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Environment

Time is discrete and infinite, $t = 0, 1, 2, \ldots$. The economy consists of a representative household, a representative competitive firm, and a benevolent government.

Household preferences

$$\sum_{t=0}^{\infty} \beta^t u(c_t, n_t), \qquad \beta \in (0, 1),$$

Production technology

$$y_t = F(K_t, n_t).$$

Environment

Resources

$$c_t + x_t + d_t = F(K_t, n_t).$$

Defense technology Define $S(d_t) \in [0, 1]$ as the fraction of exposed capital that survives when the government spends d_t on defense, with

$$S'(d) > 0, \qquad S''(d) \leq 0,$$

Law of motion of capital

$$K_{t+1} = S(d_t)\left[(1-\delta)K_t + x_t\right].$$

d_t is chosen before destruction occurs and determines the effective capital carried into period $t+1$.

Government instruments

$$\tau_t : \text{labor income tax},$$

$$\theta_t : \text{capital income tax}.$$

The first-best benchmark

Given K_0, a benevolent planner directly chooses the allocation $\{c_t, n_t, x_t, d_t\}_{t=0}^{\infty}$ to solve

$$\max \sum_{t=0}^{\infty} \beta^t u(c_t, n_t)$$

subject to, for every t,

$$c_t + x_t + d_t = F(K_t, n_t),$$

$$K_{t+1} = S(d_t)\left[(1-\delta)K_t + x_t\right].$$

FOCs:

$$x_t: \quad \underbrace{u_{c,t}}_{\text{marginal cost}} = \underbrace{\beta u_{c,t+1}\left[1 + F_{K,t+1} - \delta\right] S(d_t)}_{\substack{\text{effective marginal return} \\ \text{on investment}}},$$

$$d_t: \quad \underbrace{u_{c,t}}_{\text{marginal cost}} = \underbrace{\beta u_{c,t+1}\left[1 + F_{K,t+1} - \delta\right] S'(d_t)\left[(1-\delta)K_t + x_t\right]}_{\substack{\text{marginal return} \\ \text{on defense}}}.$$

c_t, n_t : standard

First-best investment and defense

It is optimal to invest in x and d up to the point where their marginal product are equalized at each date:

$$S(d^*) = [(1-\delta)K + x^*]S'(d^*).$$

Equivalently,

$$x^*(d;K) = \frac{S(d)}{S'(d)} - (1-\delta)K.$$

Since $S'(d) > 0$ and $S''(d) \leq 0$,

$$\frac{\partial x^*(d;K)}{\partial d} = 1 - \frac{S(d)S''(d)}{[S'(d)]^2} > 1 > 0.$$

$\Rightarrow$ For a given inherited capital stock, investment and defense are complements. When you have more capital (higher stake), defense is more valuable; more defense raises the effective return to investment.

Functional forms

Preferences

$$u(c,n) = \frac{c^{1-\sigma}-1}{1-\sigma} - \chi \frac{n^{1+1/\phi}}{1+1/\phi}.$$

Production

$$F(K,n) = K^{\alpha} n^{1-\alpha}.$$

Capital survival

$$S(d) = 1 - ae^{-bd}, \qquad a \in [0,1], \quad b > 0.$$

- a: destruction exposure.
- b: effectiveness of defense.

The survival technology satisfies

$$S(0) = 1 - a, \qquad \lim_{d\to\infty} S(d) = 1,$$

$$S'(d) = abe^{-bd} > 0, \qquad S''(d) = -ab^2 e^{-bd} < 0.$$

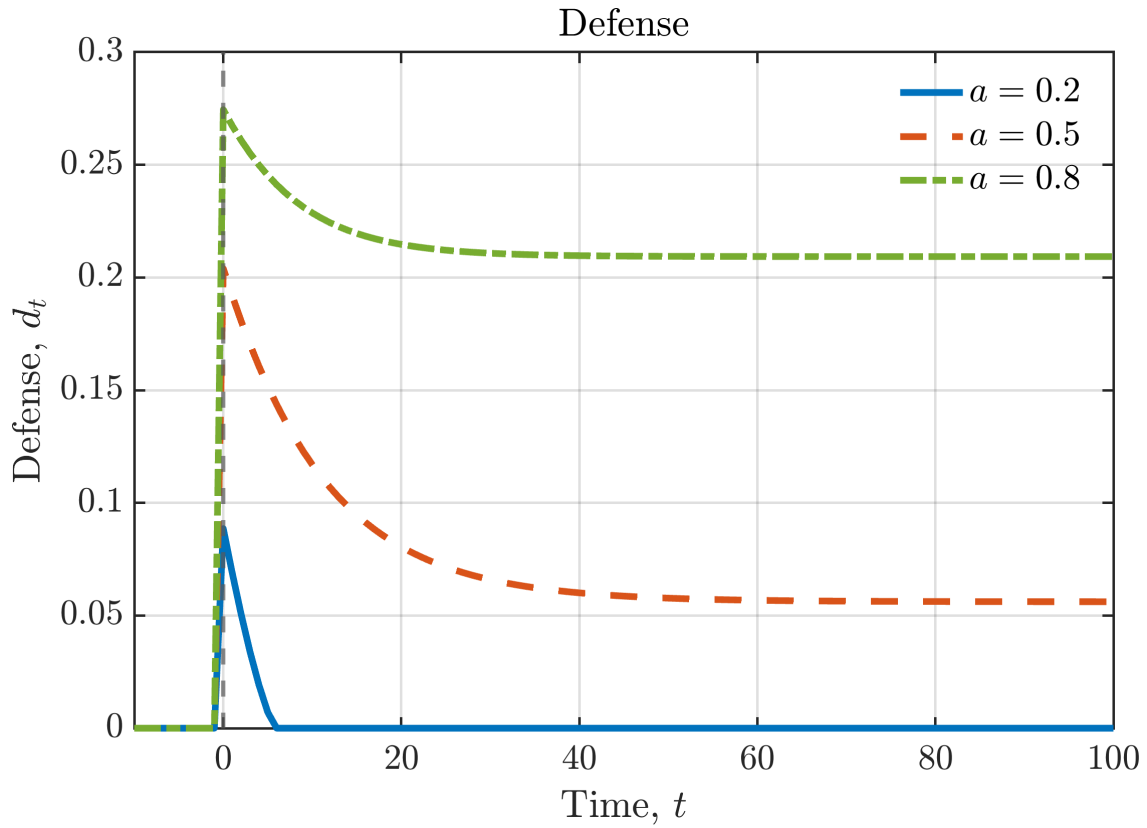
Parameters

Parameter	Value	Description	Source / Target
β	0.96	Discount factor	Annual frequency
α	0.33	Capital share	Literature
δ	0.10	Capital depreciation rate	Literature
σ	1.50	Relative risk aversion	Literature
ϕ	1.00	Frisch elasticity	Literature
χ	12.62	Labor-disutility weight	$n^{SS} = 1/3$ in no threat state
a	0.50	Baseline destruction exposure	Illustrative benchmark
b	8.00	Baseline effectiveness of defense	Illustrative benchmark

Note: Parameters a and b are illustrative and are varied in the comparative-static exercises.

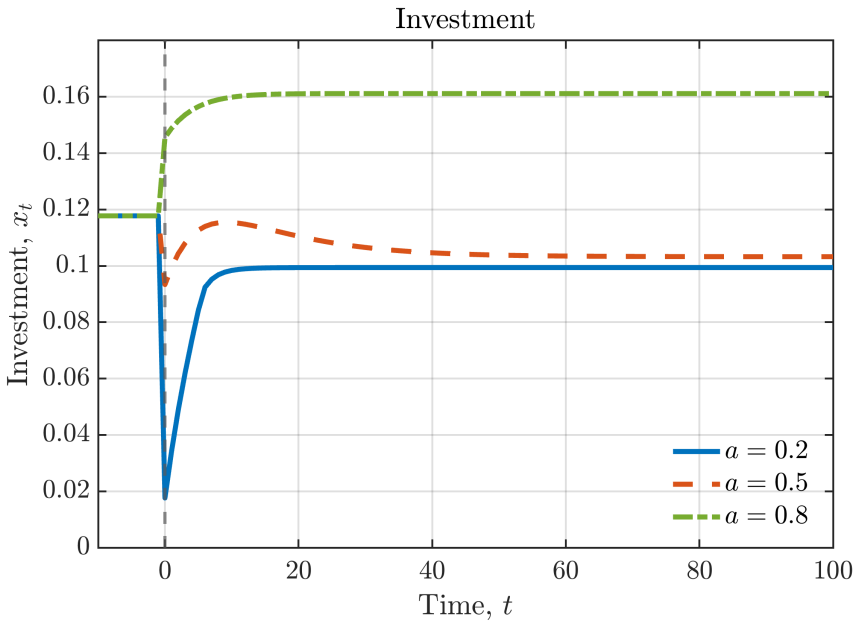
How defense responds to greater destruction exposure?

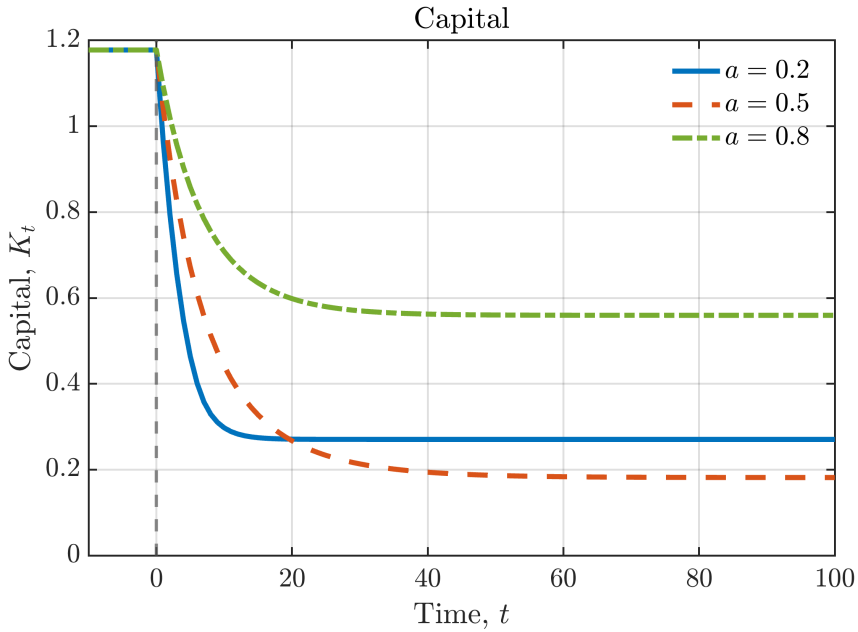
At $t = 0$, the economy experiences a permanent increase in destruction exposure: $a = 0 \to a > 0$



Onset: high initial capital and front-load $\Rightarrow$ $d \uparrow$; higher a $\to$ MP_d higher $\to$ d higher
Rebuild: lower capital exposed $\Rightarrow$ $d \downarrow$
No defense: low exposure, d limited mitigation; also compete with x.

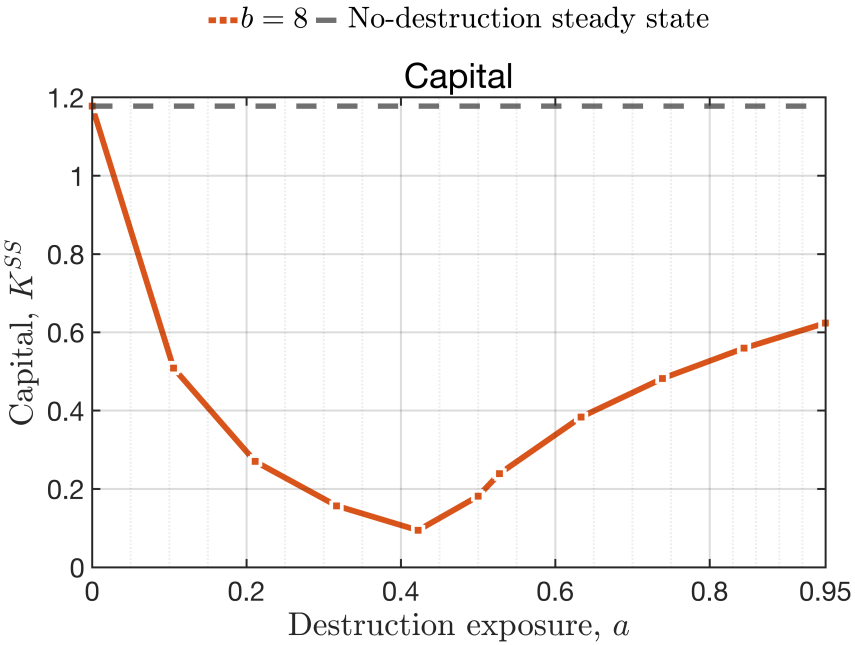
Investment and capital

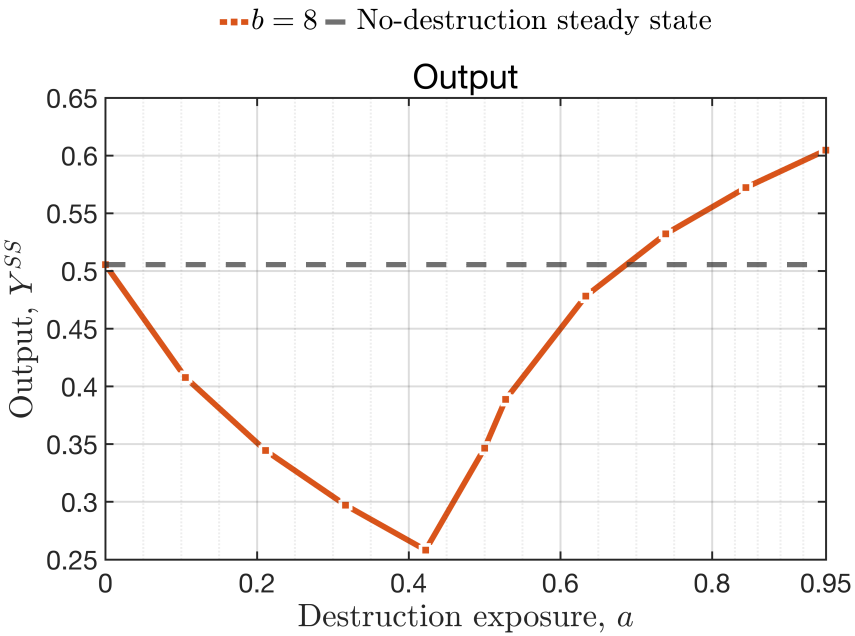




Onset: x drops due to compete with d.
Rebuild: as d is higher, $MR_x \uparrow \rightarrow x \uparrow$
Capital: in high exposure, strong d and x make K higher, partially offset destruction effect.

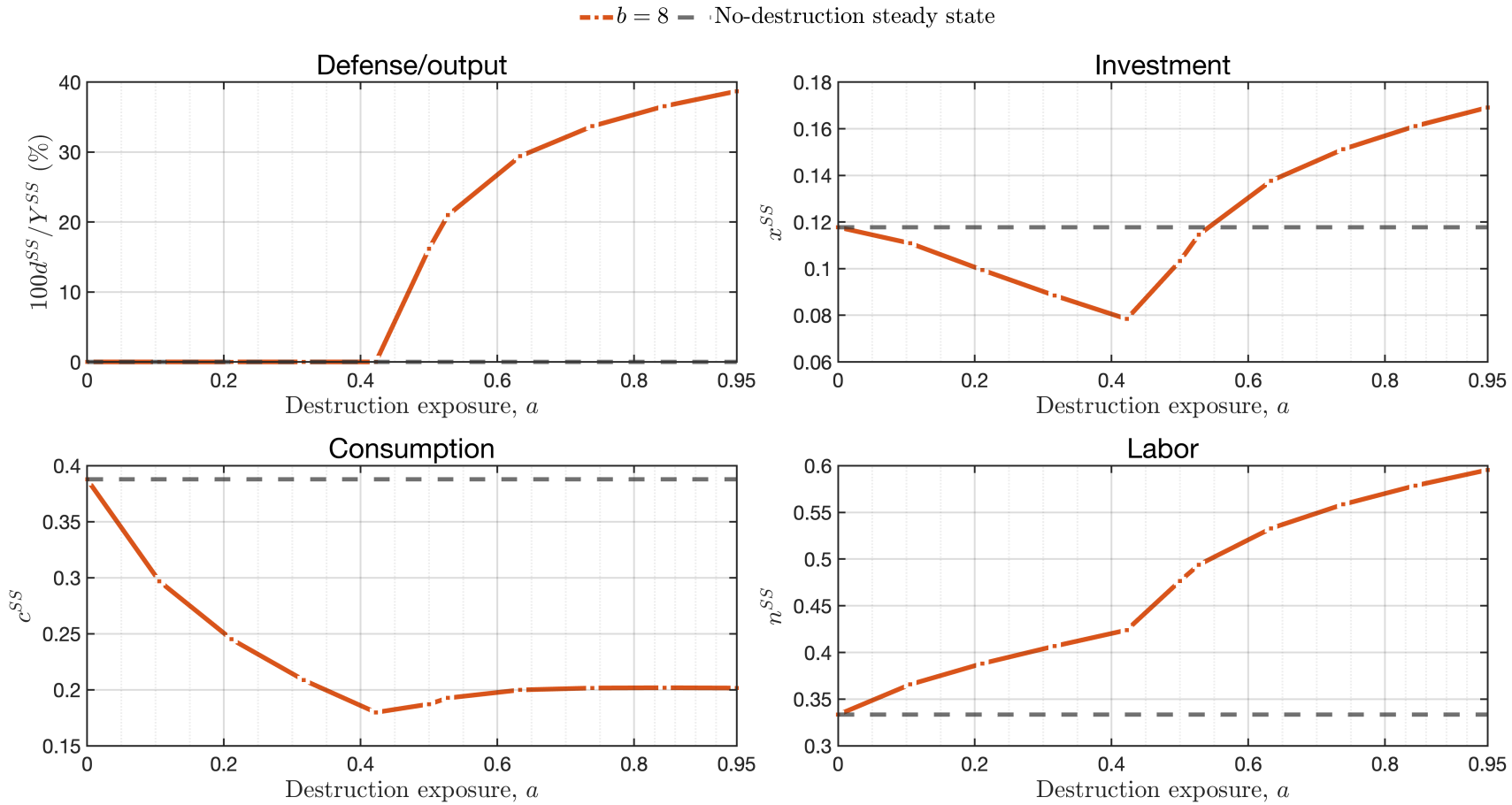
Effective defense can restore productive capacity





- Below a destruction threshold, $d^{ss} = 0$.
- Above that threshold, effective defense allows steady-state capital and output to recover.

Adaptation is costly

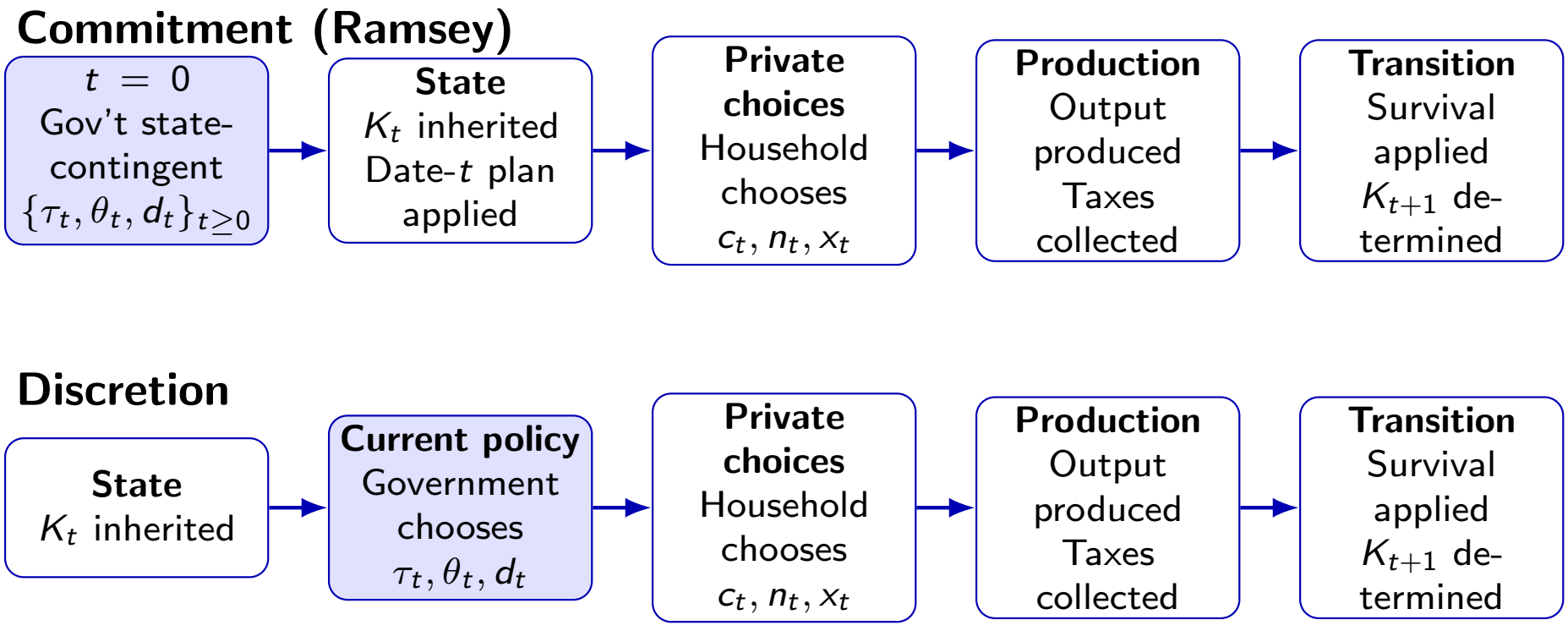


Adaptation requires more labor and limited consumption, so welfare remains below its no-destruction level.

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Timing under commitment and discretion



Key distinction: Commitment fixes the entire policy sequence at $t = 0$; under discretion, policy is reoptimized after capital has been installed.

The Ramsey problem

At $t = 0$, the government commits to a fiscal plan

$$\pi = \{\tau_t, \theta_t, d_t\}_{t=0}^{\infty}$$

and selects the plan whose induced competitive equilibrium maximizes household welfare:

$$\max_{\pi} \sum_{t=0}^{\infty} \beta^t u(c_t, n_t).$$

The induced allocation and prices must form a competitive equilibrium:

- households and firms optimize given π;
- markets clear and allocations are feasible;
- defense satisfies the government budget:

Policy is the government's choice; allocations are equilibrium responses.

The Competitive Equilibrium

The household.

The household, taking prices $\{w_t, r_t\}_{t=0}^{\infty}$ and government policies $\{\tau_t, \theta_t, d_t\}_{t=0}^{\infty}$ as given, solves

$$\max_{\{c_t, n_t, x_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t, n_t)$$

subject to, for every $t > 0$,

$$\begin{aligned} c_0 + x_0 &= (1-\tau_0) w_0 n_0 + r_0 K_0, \\ c_t + x_t &= (1-\tau_t) w_t n_t + (1-\theta_t) r_t K_t, \\ K_{t+1} &= S(d_t)\big[(1-\delta)K_t + x_t\big], \qquad K_0 \text{ given.} \end{aligned}$$

Household FOCs imply

$$\begin{aligned} -u_{n,t} &= u_{c,t}(1-\tau_t) w_t, \\ u_{c,t} &= \beta u_{c,t+1} S(d_t)\left[(1-\delta) + (1-\theta_{t+1}) r_{t+1}\right]. \end{aligned} \tag{1}$$

[1] it is typically assumed that Ramsey planner can't tax initial wealth

The Competitive Equilibrium

The competitive firm

$$w_t = F_n(K_t, n_t),$$
$$r_t = F_K(K_t, n_t).$$

Government

$$d_t = \tau_t w_t n_t + \theta_t r_t K_t$$

Markets clear

$$c_t + x_t + d_t = F(K_t, n_t)$$

An equivalent primal Ramsey problem

Household optimality allows taxes and prices to be eliminated. The resulting implementability condition is

$$\beta u_{c,t}\left[c_t + x_t + (1-\delta)K_t\right] + \beta u_{n,t} n_t = u_{c,t-1}\left[(1-\delta)K_{t-1} + x_{t-1}\right]. \qquad \text{(IC)}$$

The Ramsey government can therefore choose allocations directly:

$$\max_{\{c_t, n_t, x_t, d_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t, n_t)$$

subject to

$$\begin{aligned} c_t + x_t + d_t &= F(K_t, n_t), \\ K_{t+1} &= S(d_t)\left[(1-\delta)K_t + x_t\right], \\ (\mathrm{IC})_t &\quad \text{for all } t \geq 1. \\ (\mathrm{IC})_0 &\quad \text{for } t = 0. \end{aligned}$$

▸ Derive IC ▸ Recursive format ◂ Ramsey dual

Taxes are recovered afterward. ▸ Tax recovery

Discretion

The household takes current policy and future policy rules as given:

$$\tau_t = T(K_t), \qquad \theta_t = \Theta(K_t), \qquad d_t = D(K_t).$$

It chooses $\{c_t, n_t, x_t\}_{t=0}^{\infty}$ to maximize

$$\sum_{t=0}^{\infty} \beta^t u(c_t, n_t)$$

subject to

$$\begin{aligned} c_t + x_t &= \big[1 - T(K_t)\big] w_t n_t + \big[1 - \Theta(K_t)\big] r_t K_t, \\ K_{t+1} &= S\big(D(K_t)\big)\big[(1-\delta)K_t + x_t\big], \qquad K_0 \text{ given.} \end{aligned}$$

The household's optimality conditions are

$$\begin{aligned} -u_n(c_t, n_t) &= u_c(c_t, n_t)\big[1 - T(K_t)\big] F_n(K_t, n_t), \\ u_c(c_t, n_t) &= \beta u_c\big(C(K_{t+1}), N(K_{t+1})\big)\, S\big(D(K_t)\big) \\ &\quad * \big[1 - \delta + \big(1 - \Theta(K_{t+1})\big) F_K(K_{t+1}, N(K_{t+1}))\big]. \end{aligned}$$

The government problem under discretion

The government solves

$$V^D(K) = \max_{c,n,x,d,\tau,\theta} \left\{ u(c,n) + \beta V^D(K') \right\}$$

subject to

$$\begin{aligned}
c + x + d &= F(K,n), \\
K' &= S(d)\big[(1-\delta)K + x\big], \\
-u_n(c,n) &= u_c(c,n)(1-\tau)F_n(K,n), \\
u_c(c,n) &= \beta u_c\big(C(K'),N(K')\big)\, S(d) \\
&\quad * \left[(1-\delta) + (1-\Theta(K'))F_K\big(K',N(K')\big)\right], \\
d &= \tau F_n(K,n)n + \theta F_K(K,n)K.
\end{aligned}$$

Classic time inconsistency

Ramsey: At $t = 0$, the Ramsey government recognizes that future capital taxation affects investment today:

$$\theta_t \text{ higher} \quad \Longrightarrow \quad \text{after-tax return lower} \quad \Longrightarrow \quad x_{t-1} \text{ lower.}$$

It therefore **restrains** future capital taxation to support investment.

When period t arrives: The investment decision has already been made and K_t is installed. Government has an incentive to tax capital as it is inelastically provided.

Time inconsistency: A low/zero capital-tax policy that is promised before may not be optimal to implement after capital is installed.

Discretion: tend to tax capital.

Mechanism

Ramsey government taxes labor income:

$$\underbrace{\tau_t \uparrow \Longrightarrow \; d_t \uparrow \Longrightarrow \; S(d_t) \uparrow \Longrightarrow \; K_{t+1} \uparrow}_{\text{Protection}}$$

Discretionary government taxes capital:

$$\underbrace{\theta_t \uparrow \Longrightarrow \; d_t \uparrow \Longrightarrow \; S(d_t) \uparrow \Longrightarrow \; K_{t+1} \uparrow}_{\text{Protection}}$$

but,

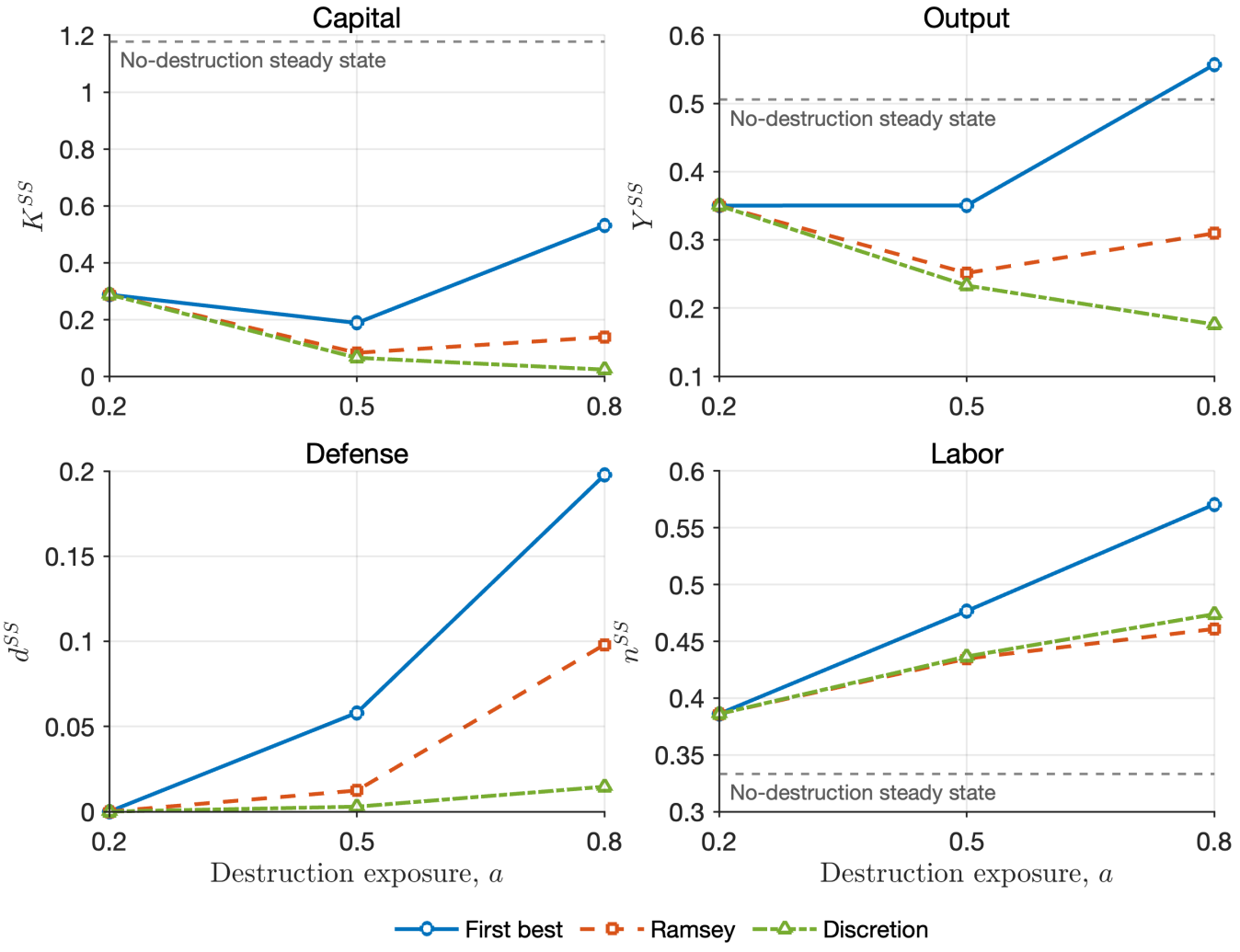
$$\underbrace{\theta_t \uparrow \Longrightarrow \; \text{after-tax return} \downarrow \Longrightarrow \; x_{t-1} \downarrow \Longrightarrow \; K_t, \text{further } K_{t+1} \downarrow}_{\text{Classic investment-distortion}}$$

$$\underbrace{\text{tax base} \downarrow \Longrightarrow \; \text{defense} \downarrow \Longrightarrow \; \text{survival} \downarrow \Longrightarrow \; K_{t+1} \text{ even} \; \downarrow}_{\text{Additional distortion}}$$

Outline

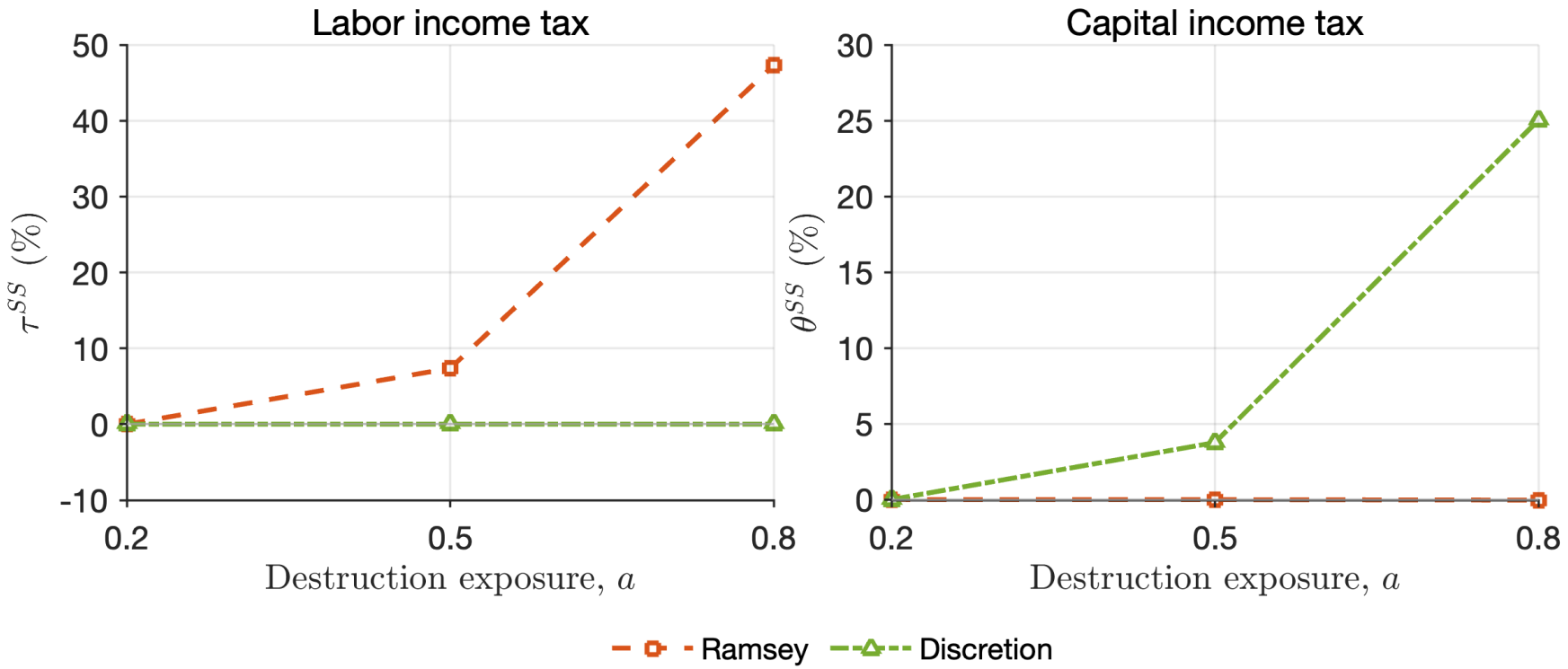
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Different responses to rising destruction exposure



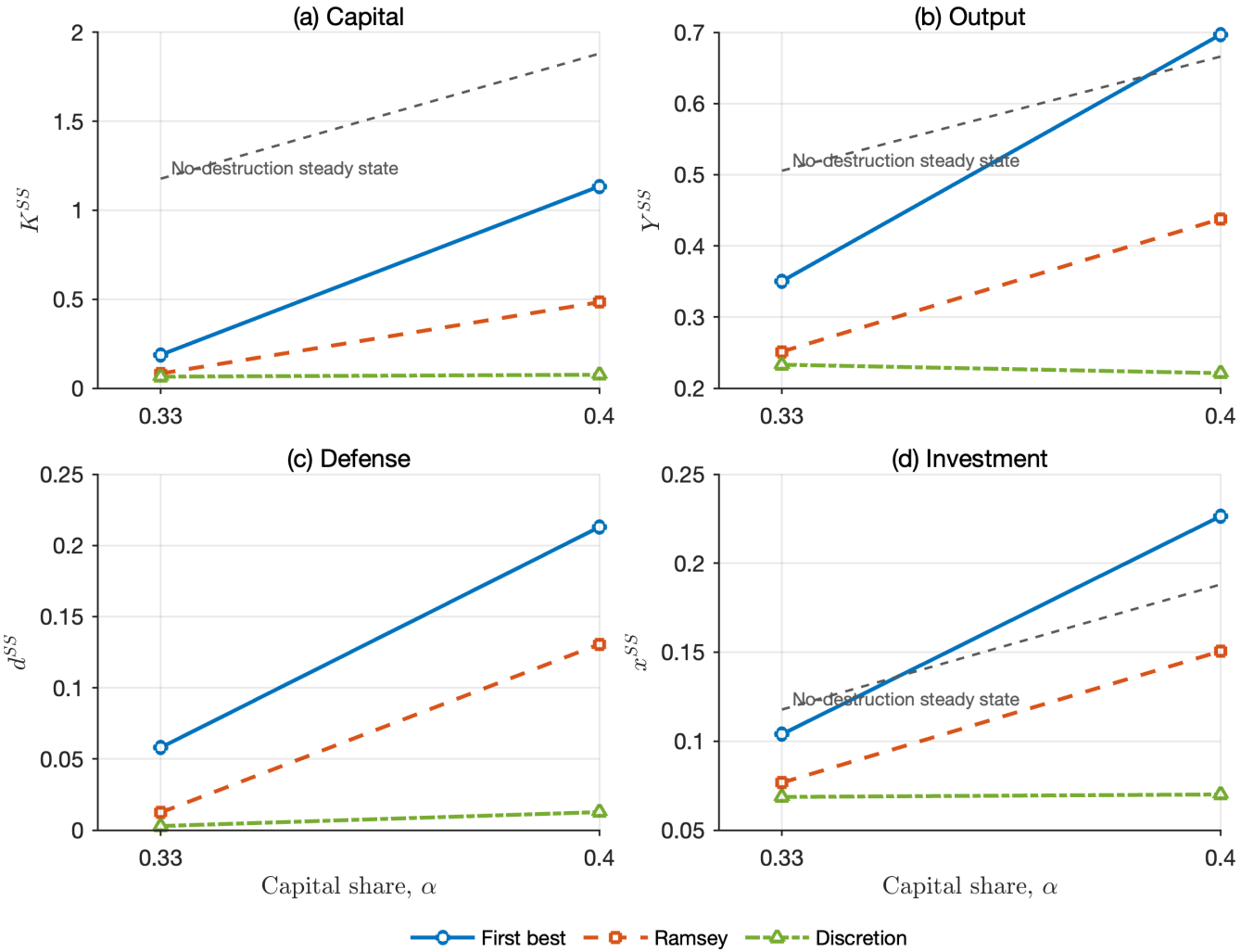
1. Distortionary taxes imply underprovision.
2. Ramsey partially adapt, but remain below the first best.
3. Under discretion: low investment, capital, and protection reinforce one another.

Tax structure



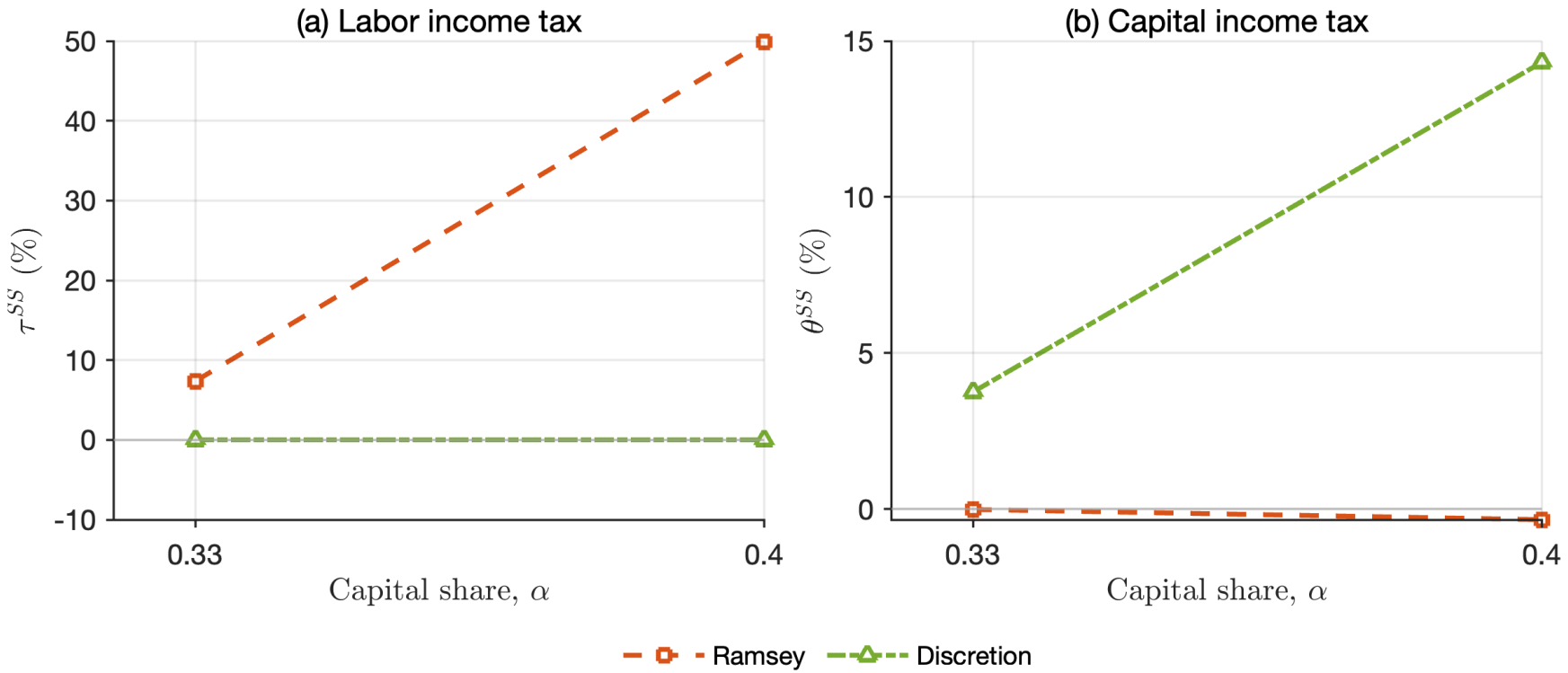
Ramsey relies primarily on labor taxation; discretion continues to exploit installed capital.

Capital intensity increases accumulation and protection



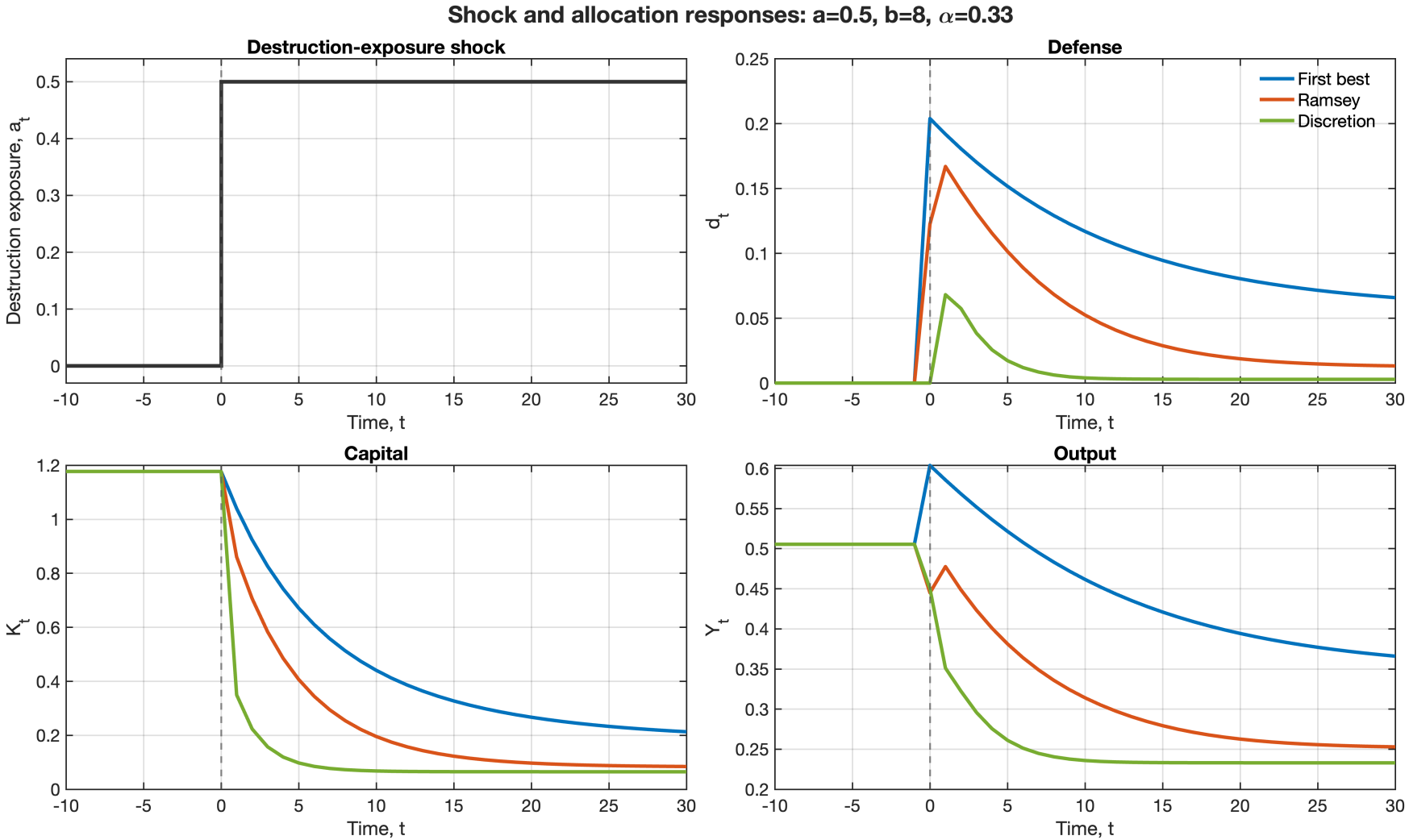
$F(K, n) = K^{\alpha} n^{1-\alpha}$. Capital becomes more valuable to protect

Ramsey subsidizes capital when capital is especially productive



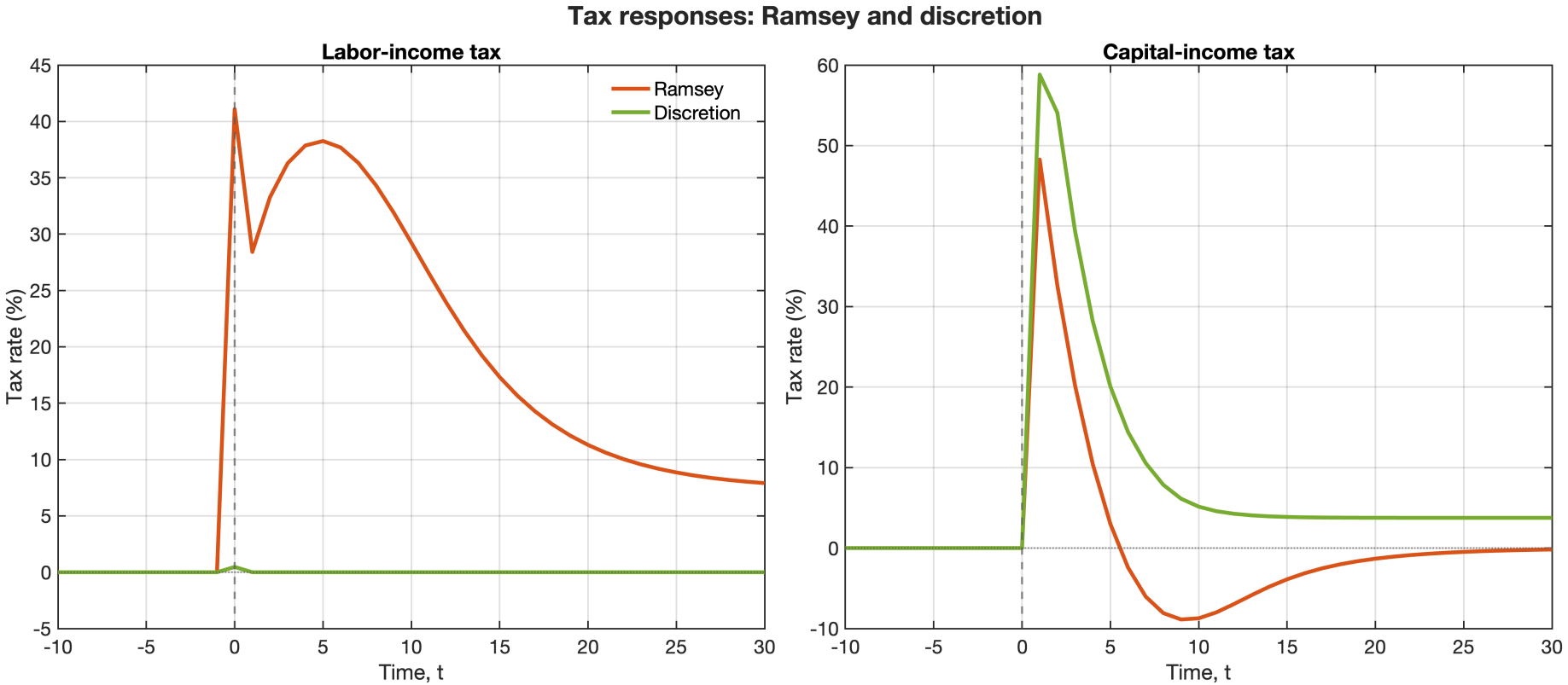
Capital becomes more valuable to protect; at high α, Ramsey may subsidize accumulation.

Transition Dynamics



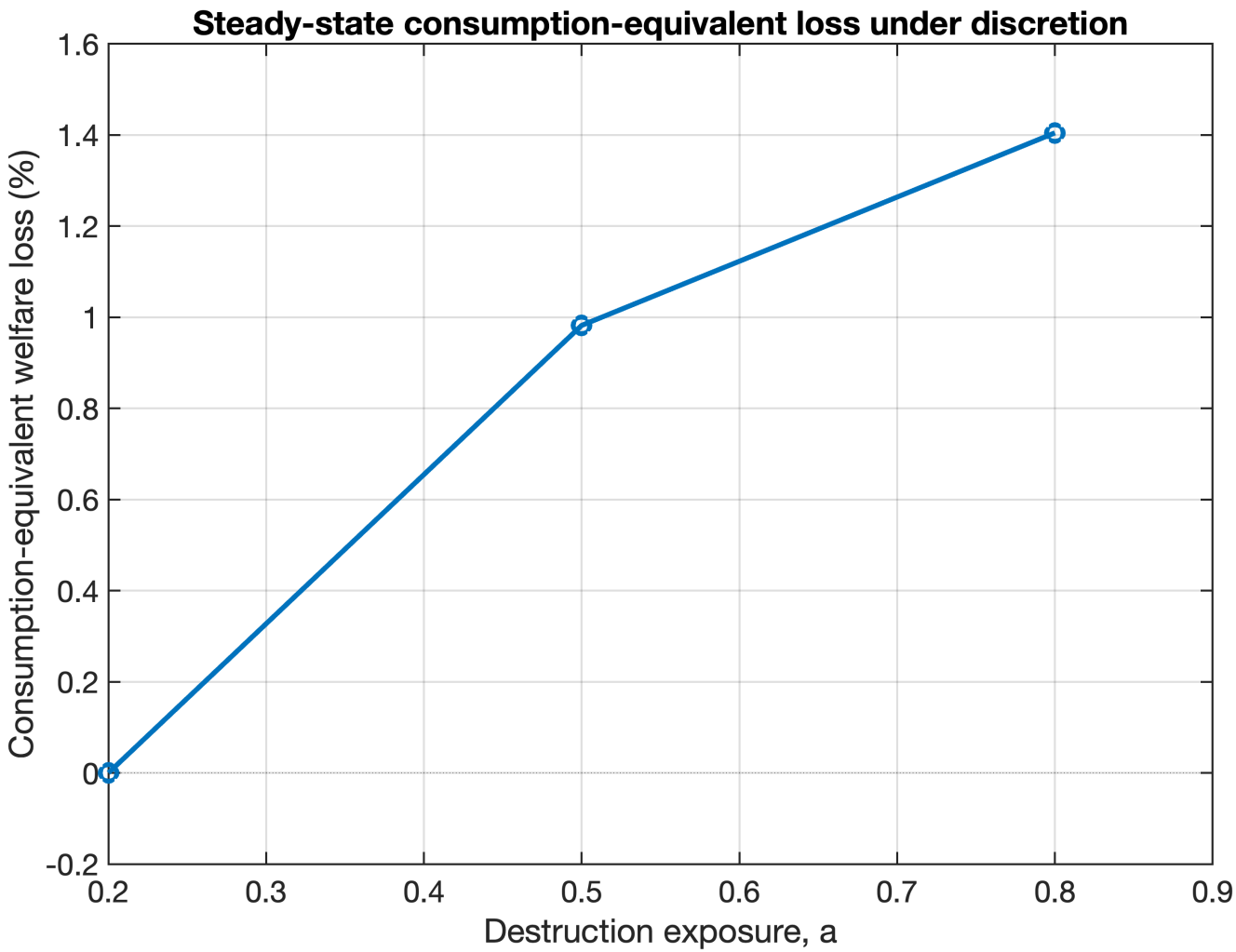
After a permanent exposure shock, discretion loses capital fastest. Weak protection and collapsed investment reinforce one another under discretion.

Ramsey protects the rebuilding margin; discretion taxes it



Ramsey temporarily subsidizes capital during adjustment; discretion raises most revenue from installed capital.

Welfare difference



At intermediate exposure, discretion generates a welfare loss of approximately **1%** relative to Ramsey commitment. The loss increases with destruction exposure.

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Back to empirical

Until now, destruction was deterministic. I now let the geopolitical threat be z_t, and it follows

$$z_{t+1} = \rho z_t + \varepsilon_{t+1}, \qquad E[\varepsilon_{t+1}] = 0.$$

The logistic transformation maps threat into a destruction probability:

$$\varphi_t = \Lambda(z_t) = \frac{1}{1 + \exp(-z_t)} \in (0, 1).$$

Let

$$K_{t+1} = \begin{cases} (1-\delta)K_t + x_t, & \text{w.p. } 1 - \varphi_t, \\ S(d_t)\big[(1-\delta)K_t + x_t\big], & \text{w.p. } \varphi_t. \end{cases}$$

$$z_t \uparrow \implies \varphi_t \uparrow \implies \text{destruction becomes more likely.}$$

The stochastic first-best problem

The planner observes (K_t, z_t) and chooses (c_t, n_t, x_t, d_t) before the destruction outcome is realized.

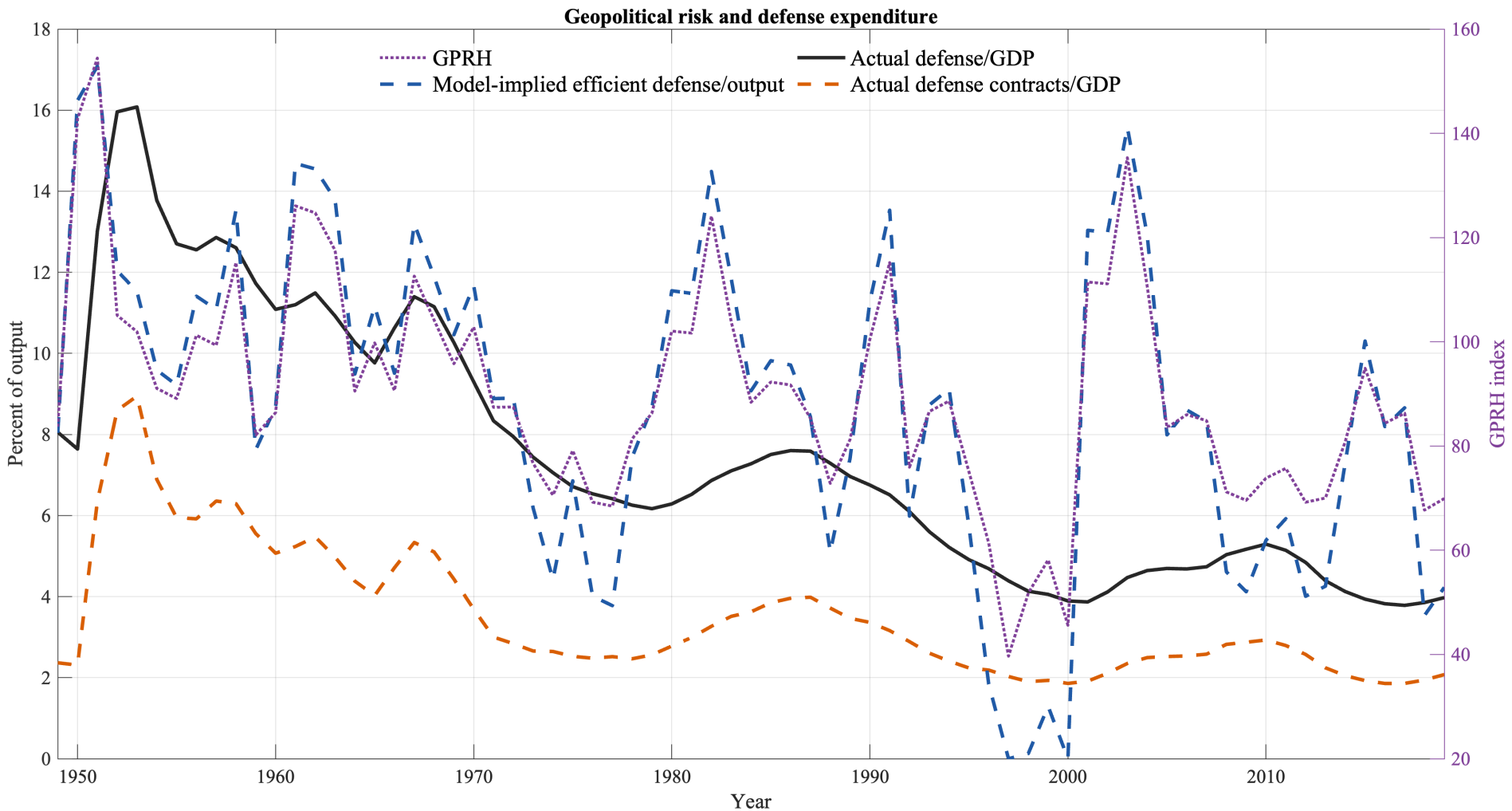
$$V^{FB}(K,z) = \max_{c,n,x,d} \Bigg\{ u(c,n) + \beta E_{z'|z} \Big[(1-\varphi) V^{FB}\big((1-\delta)K + x, z'\big) + \varphi V^{FB}\Big(S(d)\big[(1-\delta)K + x\big], z'\Big)\Big] \Bigg\}$$

subject to

$$c + x + d = F(K,n),$$
$$z' = \rho z + \varepsilon',$$
$$\varphi = \frac{1}{1+\exp(-z)}.$$

▸ Mapping GPR

Model-implied and actual defense expenditure



Same response direction; but more volatile as actual spending reflects additional forces: standing military commitments, procurement programs, political constraints etc.

Summary

- Optimal defense spending increases with destruction exposure.
- An economy can adapt to riskier environment by diverting resources optimally and reach a higher level of capital stock and output; while welfare continues to fall.
- Under Ramsey, defense is financed mainly through labor taxes, and capital may be subsidized during rebuilding; increase the economy's ability to adapt.
- Under discretion, installed capital remains an attractive tax base. Capital taxation reduces investment, the capital stock and ultimately protection.
- The welfare gain from policy credibility increases with destruction risk.
- Applications beyond defense: cybersecurity (public cyber defense protects digital infrastructure from disruption or destruction.)
- Stochastic risk extension in Ramsey and discretion.

Ramsey problem

The optimal taxation problem is then to choose a fiscal plan $\{\pi_t = (\tau_t, \theta_t, d_t)\}_{t=0}^{\infty}$ at time 0, anticipating the competitive-equilibrium allocation it induces.

It solves

$$\max_{\{\pi_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t, n_t)$$

st.

$$\begin{aligned}
c_t + x_t + d_t &= F(K_t, n_t), && \text{Resource constraint} \\
K_{t+1} &= S(d_t)\left[(1-\delta)K_t + x_t\right], && \text{Capital accumulation} \\
-u_{n,t} &= u_{c,t}(1-\tau_t)w_t, && \text{Labor supply} \\
u_{c,t} &= \beta u_{c,t+1} S(d_t)\left[(1-\delta) + (1-\theta_{t+1})r_{t+1}\right], && \text{Euler equation} \\
w_t &= F_n(K_t, n_t), \qquad r_t = F_K(K_t, n_t), && \text{Factor prices} \\
d_t &= \tau_t w_t n_t + \theta_t r_t K_t, && \text{Government budget}
\end{aligned}$$

◂ ramsey primal

The Primal Approach

Instead of choosing taxes and solving for the resulting equilibrium, the Ramsey planner chooses allocations directly:

$$\mathcal{A} = \{c_t, n_t, x_t, d_t\}_{t=0}^{\infty}.$$

The allocation must satisfy two requirements:

1. **Feasibility**

   The resource constraint and capital law of motion hold.

2. **Implementability**

   There exists a sequence of taxes $\{\tau_t, \theta_t\}_{t=0}^{\infty}$ that supports $\mathcal{A}$ as a competitive equilibrium.

Procedure: Solve for the optimal implementable allocation first, then back out the tax rates from the household optimality conditions.

Ramsey chooses the best implementable allocation

Using the household labor condition, the capital Euler equation, and the household budget constraint, we can eliminate τ_t and θ_t. This gives the implementability condition (IC)

$$\beta u_{c,t}\left[c_t + x_t + (1-\delta)K_t\right] + \beta u_{n,t} n_t = u_{c,t-1}\left[(1-\delta)K_{t-1} + x_{t-1}\right]. \qquad \text{(IC)}$$

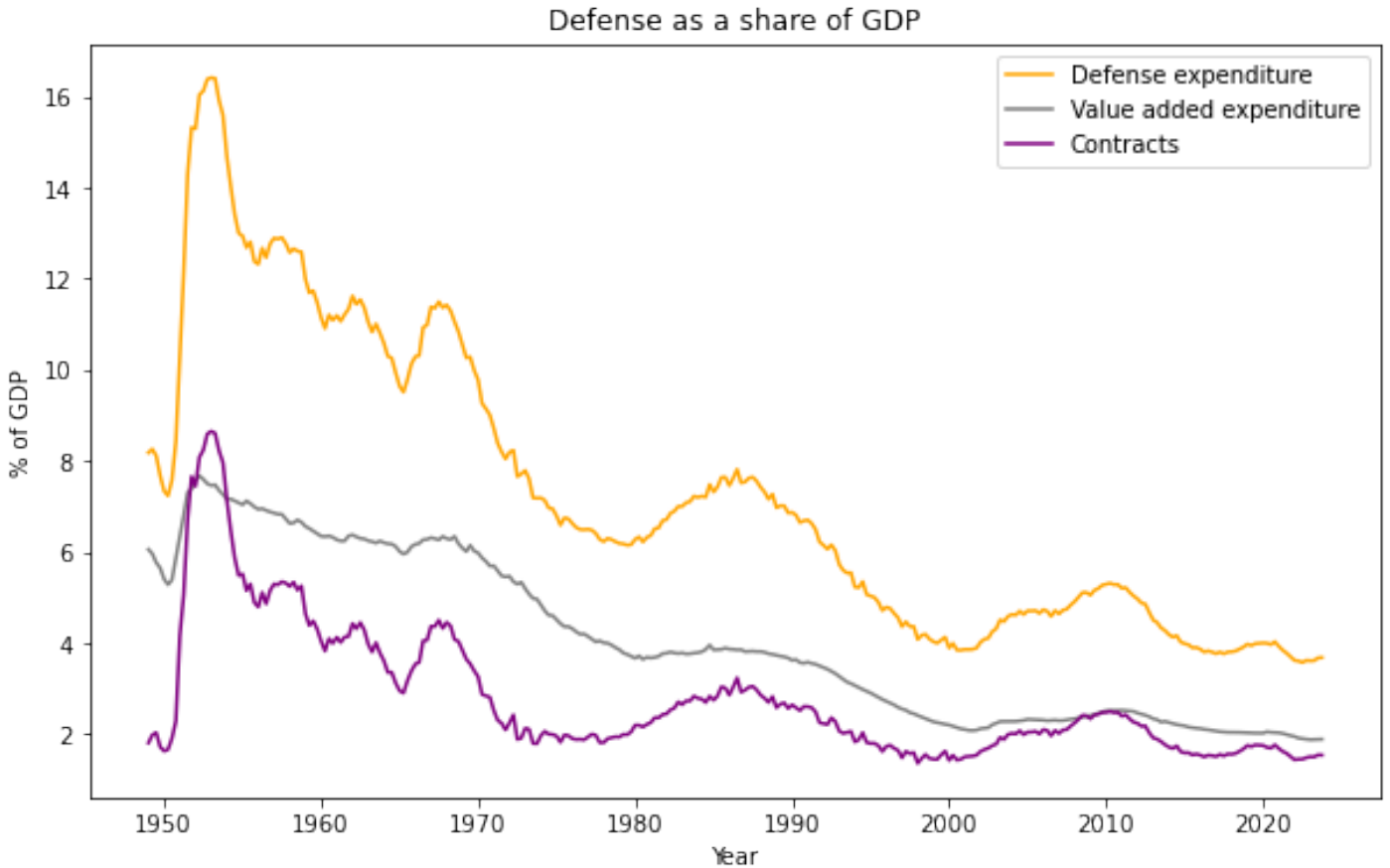
The Ramsey problem can be written in primal form:

$$\max_{\{c_t, n_t, x_t, d_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t, n_t)$$

subject to

$$\begin{aligned} c_t + x_t + d_t &= F(K_t, n_t), \\ K_{t+1} &= S(d_t)\left[(1-\delta)K_t + x_t\right], \\ (\mathrm{IC})_t \qquad & \text{for all } t \geq 1. \end{aligned}$$

U.S Government Spending Decomposition



- Defense contracts dominates the variation in defense spending.

◂ Return to problem

Contribution of each component to variation in big G

Table: Contribution to Variation in G

			Contribution to R^2	Weight in G
Federal			90%	40%
	Defense		78%	24%
		Df Contracts	65%	10%
	Nondefense		12%	16%
State&Local			10%	60%

◂ Return to problem

Deriving the implementability condition

The household budget constraint and optimality conditions are

$$c_t + x_t = (1-\tau_t)w_t n_t + (1-\theta_t)r_t K_t, \tag{HB}$$

$$u_{n,t} + u_{c,t}(1-\tau_t)w_t = 0, \tag{LS}$$

$$u_{c,t-1} = \beta u_{c,t} S(d_{t-1}) \left[(1-\theta_t)r_t + 1 - \delta\right]. \tag{EE}$$

Using (LS) in (HB) eliminates the labor tax:

$$u_{c,t}(c_t + x_t) + u_{n,t} n_t = u_{c,t}(1-\theta_t)r_t K_t. \tag{1}$$

Multiply (EE) by $[(1-\delta)K_{t-1} + x_{t-1}]$ and use

$$K_t = S(d_{t-1})\left[(1-\delta)K_{t-1} + x_{t-1}\right].$$

Then

$$u_{c,t-1}\left[(1-\delta)K_{t-1} + x_{t-1}\right] = \beta u_{c,t} K_t \left[(1-\theta_t)r_t + 1 - \delta\right].$$

Hence,

$$u_{c,t}(1-\theta_t)r_t K_t = \frac{u_{c,t-1}}{\beta}\left[(1-\delta)K_{t-1} + x_{t-1}\right] - u_{c,t}(1-\delta)K_t. \tag{2}$$

Substituting (2) into (1) and rearranging gives

$$\beta u_{c,t}\left[c_t + x_t + (1-\delta)K_t\right] + \beta u_{n,t} n_t = u_{c,t-1}\left[(1-\delta)K_{t-1} + x_{t-1}\right] \tag{IC}$$

◂ Return to problem

The Ramsey problem can be written recursively

Past household decisions are summarized by the inherited state

$$q_t \equiv u_{c,t-1}\left[(1-\delta)K_{t-1} + x_{t-1}\right].$$

The recursive Ramsey problem is

$$V^R(K,q) = \max_{c,n,x,d} \left\{u(c,n) + \beta V^R(K',q')\right\}$$

subject to

$$\begin{aligned} c + x + d &= F(K,n), \\ K' &= S(d)\left[(1-\delta)K_t + x_t\right], \\ \beta u_c(c,n)(c + x + (1-\delta)K) + \beta u_n(c,n)n &= q, \\ q' &= u_c(c,n)\left[(1-\delta)K_t + x_t\right]. \end{aligned}$$

◂ Return to problem

Recovering the Ramsey taxes

The labor tax follows from the intratemporal condition:

$$\tau_t = 1 + \frac{u_{n,t}}{u_{c,t} F_n(K_t, n_t)}.$$

For $t \geq 1$, the capital tax follows from the Euler equation:

$$\theta_t = 1 - \frac{\frac{u_{c,t-1}}{\beta u_{c,t} S(d_{t-1})} - (1 - \delta)}{F_K(K_t, n_t)}.$$

The implementability condition guarantees that these taxes satisfy the government budget constraint.

◂ Return to problem

Mapping GPRH into the model threat state

I convert annual historical geopolitical risk into a standardized state:

$$GPRH_t \quad \longrightarrow \quad z_t = \frac{\log(GPRH_t) - \mu_{\log GPRH}}{\sigma_{\log GPRH}}.$$

The threat state follows a persistent process:

$$z_{t+1} = 0.864\, z_t + \varepsilon_{t+1}, \qquad \sigma_\varepsilon = 0.507.$$

I then map the unbounded state into a number between zero and one:

$$\varphi_t = \frac{1}{1 + \exp(-z_t)}.$$

For illustration,

z_t	-1	0	1
φ_t	0.269	0.500	0.731

$$GPRH_t \uparrow \quad \Longrightarrow \quad z_t \uparrow \quad \Longrightarrow \quad \varphi_t \uparrow .$$

φ_t is a normalized model probability, not an observed annual frequency of war.

◂ Return to stochastic FB